

SMART RETAIL DEMAND FORECASTING SYSTEM

PROJECT REPORT

Submitted By

ABHISHEK NARAYANAN P S (TKM22CS007),
AFREEN SUDHEER (TKM22CS014),
JENNA NAUREEN (TKM22CS080),
KEVIN SKARIAH BIJU (TKM22CS085),
NITHIN M (LTKM22CS146)

to

TKM College of Engineering
in partial fulfilment of the requirements for the award of
B.Tech Degree in Computer Science and Engineering



Department of Computer Science and Engineering
TKM College of Engineering, Kollam, Kerala, 691005,
April 2026

DECLARATION

We, the undersigned, hereby declare that the report on "**SMART RETAIL DEMAND FORECASTING SYSTEM**," submitted as part of the course Computer Science Engineering, under TKM College of Engineering, is a bonafide work done by us under the supervision of Dr. Reshma Sheik, Assistant Professor, Department of Computer Science and Engineering.

This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately cited and referenced the sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in our submission.

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Place: Kollam
Date: 13/04/2026

ABHISHEK NARAYANAN P S
AFREEN SUDHEER
JENNA NAUREEN
KEVIN SKARIAH BIJU
NITHIN M

Department of Computer Science and Engineering
TKM College of Engineering, Kollam



CERTIFICATE

This is to certify that the report titled "**SMART RETAIL DEMAND FORECASTING SYSTEM**" submitted by **ABHISHEK NARAYANAN PS (TKM22CS007)**, **AFREEN SUDHEER (TKM22CS014)**, **JENNA NAUREEN (TKM22CS080)**, **KEVIN SKARIAH BIJU (TKM22CS085)**, and **NITHIN M (LTKM22CS146)** to **TKM College of Engineering** in partial fulfillment of the requirements for the award of B.Tech Degree in Computer Science and Engineering during 2025-26 is a bonafide record of the project work carried out by them under our guidance and supervision.

Internal Supervisor	Project Coordinator	Head of Department	External Examiner
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ACKNOWLEDGEMENT

We take this opportunity to express our deep sense of gratitude to the Almighty and sincere thanks to all who helped us to complete the project successfully.

We express our sincere gratitude to **Dr. Sadiq A**, Principal, TKMCE, for providing us with all necessary facilities and support for the project work. We are extremely grateful to **Dr. Aneesh G Nath**, Head of Department, Department of Computer Science and Engineering, and our project guide, **Dr. Reshma Sheik**, Assistant Professor, for their constructive guidance, advice, constant support, and timely suggestions; this project would not have been possible.

We extend our immense gratitude to all faculty and technical staff in the Computer Science department for their help and necessary facilities. Our humble gratitude and heartiest thanks also go to our parents and friends who have supported and helped us throughout this work.

Place: Kollam
Date: 13/04/2026

ABHISHEK NARAYANAN P S
AFREEN SUDHEER
JENNA NAUREEN
KEVIN SKARIAH BIJU
NITHIN M

ABSTRACT

Effective inventory management requires accurate retail demand forecasting, yet traditional and foundation models often lag behind rapid market shifts. This project presents the Smart Retail Demand Forecasting System, a hybrid forecasting framework that enhances baseline predictions from classical (ARIMA, STL-ARIMA), time-series (Prophet), deep learning (LSTM), and foundational (Chronos) models using a model-agnostic Sales Momentum Index (SMI) post-processing layer.

SMI captures recent demand trends through three derived signals: Hype Index (spike intensity), Momentum (trend acceleration), and Persistence (trend stability), all computed directly from historical sales data. The rule-based Nudger applies these signals to adjust 12-week forecasts without model retraining, ensuring interpretability and computational efficiency.

Evaluated on the Walmart Recruiting Store Sales and M5 Forecasting datasets, the system demonstrates significant MASE and WQL improvements on structured Walmart data (Chronos and SMI: ~ 0.045 MASE reduction) while showing marginal gains on sparse M5 series where STL-ARIMA remains competitive. Results confirm SMI-nudging enhances stable time series most effectively while preserving model rankings.

This work contributes a practical, transparent enhancement framework for retail forecasting, bridging the gap between static predictions and dynamic demand patterns.

Keywords: [Retail Demand Forecasting, Chronos Foundation Model, Sales Momentum Index, Rule-Based Nudging, Time-Series Post-Processing]

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	Auto-Regressive Integrated Moving Average
CSV	Comma-Separated Values
CPI	Consumer Price Index
EMA	Exponential Moving Average
FLOPs	Floating Point Operations
IDE	Integrated Development Environment
KPIs	Key Performance Indicators
LTS	Long Term Support
LSTM	Long Short-Term Memory
ML	Machine Learning
MASE	Mean Absolute Scaled Error
PO	Program Outcome
SDGs	Sustainable Development Goals
SKUs	Stock Keeping Units
SMI	Sales Momentum Index
STL-ARIMA	Seasonal Trend decomposition using Loess-AutoRegressive Integrated Moving Average
UI	User Interface
WQL	Weighted Quantile Loss

LIST OF SYMBOLS

Symbol	Meaning
<i>hype_volatility</i>	Standard deviation of the SMI Hype Index over the last 8 weeks
<i>momentum_strength</i>	Absolute latest momentum
<i>persistence_score</i>	Average persistence
<i>base</i>	Base nudge value taken from the horizon-dependent regime
<i>confidence</i>	Confidence multiplier computed from SMI signals
<i>multiplier</i>	Final adjustment multiplier applied to the baseline forecast
<i>decay / decay(horizon)</i>	Horizon-aware decay factor function
<i>ewm</i>	Method used to provide "Exponential Weighted" (EW) functions, for calculating an Exponential Moving Average (EMA)
<i>n</i>	Total number of observations in the training period
<i>H</i>	Forecast horizon (number of steps ahead)
y_t	Actual observed value at time t
$\hat{y}_t$	Predicted value (point forecast) at time t
$\hat{y}_{t,q}$	Quantile-specific predicted value at time t
<i>m</i>	Seasonal period
<i>q</i>	Quantile level (e.g., $q = 0.1, 0.5, 0.9$)

CHAPTER 1 : INTRODUCTION

Retail demand forecasting is central to inventory planning, replenishment, and supply-chain efficiency. While classical and modern forecasting models capture long-term trend and seasonality, they often react slowly to short-term demand shocks.

In practical retail settings, this lag can increase stockouts, overstock, and planning uncertainty.

This work proposes a lightweight post-processing framework that improves forecast responsiveness without retraining baseline models.

The framework computes a *Sales Momentum Index (SMI)* directly from recent sales history and applies a bounded, interpretable rule-based *Nudger* to baseline outputs. Because the approach is model-agnostic, it can be applied uniformly to classical baselines (ARIMA, STL-ARIMA, Prophet), deep learning models (LSTM), and foundation models (Chronos).

The system is evaluated on two contrasting benchmarks: the Walmart Recruiting Store Sales dataset (structured weekly demand) and the M5 dataset (sparser, more intermittent demand). Results are reported using MASE and WQL, with additional computational metrics for deployment relevance.

1.1 Motivation

Organizations need forecasting systems that are accurate, fast to adapt, and operationally interpretable. Retraining complex models for every local demand shift is often costly and slow, especially in multi-store or multi-item environments. A practical alternative is to preserve strong baseline models and add a transparent correction layer that responds to recent momentum.

The proposed SMI–Nudger design addresses this need by:

- using only recent sales behavior (no mandatory external feeds),
- applying bounded adjustments to reduce instability,
- preserving interpretability for business users.

1.2 Problem Statement

Existing forecasting pipelines face three recurring limitations in retail deployment:

1. **Short-term lag:** forecasts do not quickly reflect sudden demand changes.
2. **Low interpretability of corrections:** many adaptive adjustments are opaque.
3. **High maintenance cost:** frequent retraining is computationally expensive.

Therefore, the core problem addressed in this work is the following:

How can baseline retail demand forecasts be improved using recent momentum signals through an interpretable, bounded, and model-agnostic post-processing layer without re-training?

1.3 Objectives

The primary goal of this project is to design and implement a hybrid retail demand forecasting system that enhances the accuracy of the predictions by combining the baseline models with a rule-based post-processing mechanism.

1. **Improve Forecast Accuracy**
 - Enhance predictions generated by classical (STL-ARIMA) and foundational (Chronos) models
 - Reduce forecasting errors such as MASE, and WQL
 - Align predictions closer to actual sales trends
2. **Develop a rule-based nudger system**
 - Design an interpretable post-processing block (Nudger)
 - Apply controlled adjustments to baseline forecasts
 - Avoid the need for retraining expensive models
3. **Utilization of Sales Momentum Index**
 - Utilize Hype Index, Persistence, and Momentum
 - Capture leading indicators of demand changes
 - Detect early signals of rising or declining demand
 - Incorporate SMI (Sales Momentum Index) Variables
4. **Enable Real Time Forecast Adaptation**
 - Use rule-based logic instead of black-box adjustments
 - Ensure transparency in how predictions are modified
 - Provide explainable forecasting outputs
5. **Ensure System Efficiency and Zero-Shot Inference**
 - Validate the system across multiple stores
 - Apply Nudger logic to unseen datasets
 - Demonstrate robustness across different retail scenarios
6. **Ensure Scalability and Generalization**
 - Reduce over-forecasting and under-forecasting
 - Improve inventory planning and supply chain efficiency

1.4 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2: Literature Review** provides an overview of the current state of retail demand forecasting, discussing traditional statistical methods, modern machine learning approaches, and the emergence of foundation models in the domain.
- **Chapter 3: Methodology** details the design and implementation of the Smart Retail Demand Forecasting System. It explains the calculation of the Sales Momentum Index (SMI), the integration of the Rule-Based Nudger, and the architectural deployment using microservices.
- **Chapter 4: Results and Experimental Analysis** presents the performance evaluation of the proposed system. It includes a comparison of the accuracy of the forecasts across different models and an analysis of the impact of the SMI-Nudger logic on inventory optimization.

- **Chapter 5: Conclusion and Future Work** summarizes the key findings of the project, discusses its practical implications for the retail industry, and identifies potential areas for further enhancement and research.

CHAPTER 2 : LITERATURE REVIEW

Retail demand forecasting has evolved through three distinct phases: classical statistical methods, machine learning approaches, and the recent emergence of foundation models. Classical techniques like ARIMA and exponential smoothing established robust benchmarks for structured retail data, while deep learning methods such as LSTM addressed non-linear patterns. Foundation models like Chronos now enable zero-shot forecasting across diverse time series. This review organizes prior work into four categories, classical forecasting baselines, deep learning and statistical models, foundation model advancements, and exogenous signal integration, before synthesizing their collective limitations that motivate our SMI-nudging framework.

2.1 Classical Forecasting Baselines

Classical time series methods form the established foundation for retail demand prediction. Kachmar et al. (2021) established ARIMA-based approaches as reliable benchmarks for store-level sales forecasting on the Walmart Recruiting dataset. Their seasonal ARIMA variants effectively captured weekly periodicities and promotional effects, achieving competitive MASE scores on store-department aggregates. The strength of the method lies in its interpretability and stability under structured seasonality.

Putri et al. (2020) extended this paradigm through exponential smoothing for periodic supply chain series. Their moving average frameworks minimized bullwhip effects by dampening demand signal amplification, demonstrating superior performance on hierarchical retail data with known seasonality. These approaches established MASE as the standard scale-independent metric for retail forecasting evaluation.

2.2 Deep Learning and Statistical Hybrids

Deep learning introduced non-linear pattern recognition to retail forecasting. Standard LSTM architectures excel at capturing long-term dependencies in sequential sales data, though they require substantial training data and suffer from vanishing gradient issues on sparse series. Prophet, developed by Facebook's Core Data Science team (Taylor & Letham, 2018), combines additive trend, seasonality, and holiday components through a generalized linear model framework. Its Bayesian uncertainty estimation proved valuable for probabilistic retail forecasts, though it struggles with sudden demand shocks absent from training data.

STL-ARIMA hybrids represent a pragmatic middle ground, decomposing series into trend, seasonal, and residual components before ARIMA fitting on residuals. This approach maintains classical interpretability while handling non-stationarity more robustly than pure ARIMA, making it particularly effective for Walmart-like datasets with clear weekly cycles.

2.3 Foundation Models and Zero-Shot Forecasting

Time series foundation models mark a paradigm shift toward scalable, pre-trained inference. Amazon's Chronos (Pineda et al., 2024) introduced the first pre-trained transformer for zero-shot time series forecasting, tokenizing historical series into discrete distributions and scaling laws from lan-

guage modeling. Applied directly to retail benchmarks, Chronos outperformed statistical baselines on structured datasets while requiring no task-specific training.

ChronosX extended this capability by incorporating exogenous covariates at inference time, demonstrating that prompting with external signals enhances zero-shot adaptation. In-context fine-tuning approaches further refined this paradigm, showing that providing related time series examples during inference dynamically adapts foundation models to target distributions without parameter updates.

2.4 Exogenous Signal Integration

Early warning signals from external sources have long augmented endogenous sales forecasting. Cao et al. (2022) developed the Hype Index for financial markets, quantifying disproportionate media attention relative to market capitalization. Their news-count-based formulation, normalizing article volume by economic size, produced bounded $[0,1]$ signals predictive of volatility and short-term movements.

Badulescu et al. (2023) extracted social media platform traces (likes, shares, early interactions) as multivariate inputs for promotional product demand modeling. These high-frequency signals captured consumer engagement momentum, improving forecast alignment during marketing campaigns. Exponential smoothing variants stabilized these noisy inputs, preserving trend memory while mitigating short-term noise amplification.

2.5 Limitations of Existing Work

Despite these advances, critical gaps persist across all categories. Classical methods like ARIMA and STL-ARIMA excel on structured data but cannot incorporate real-time momentum without exogenous regressors, limiting responsiveness to demand shocks. Deep learning approaches (LSTM, Prophet) suffer from data hunger and black-box opacity, while foundation models like Chronos provide strong zero-shot baselines yet underperform when recent demand patterns deviate from pre-training distributions.

Exogenous signal research remains domain-specific: financial Hype Index formulations do not translate to retail sales spikes, and social media traces require external feeds unavailable in most enterprise settings. No existing framework generates dataset-native momentum signals *and* applies them through a model-agnostic post-processing layer without retraining. Multi-model comparative evaluations are rare, leaving uncertainty about how momentum correction impacts classical versus foundation model rankings.

These limitations motivate our approach: deriving Sales Momentum Index signals (Hype, Momentum, Persistence) directly from sales history to create a universal nudger block. This bridges the gap between static baseline forecasts and dynamic demand patterns while maintaining full interpretability across all model families.

The surveyed literature establishes three prerequisites for our system:

- robust baselines from classical and foundation models,
- validated momentum signal construction principles, and
- the need for lightweight post-hoc enhancement.

Our methodology in Chapter 3 directly addresses these gaps through an integrated SMI-nudging pipeline evaluated across the full spectrum of forecasting paradigms.

CHAPTER 3 : METHODOLOGY

3.1 System Architecture

The proposed system follows a modular tripartite architecture comprising baseline forecasting models, the Sales Momentum Index (SMI) signal generator, and the rule-based Nudger post-processing block. Baseline forecasts are generated independently by classical (ARIMA, STL-ARIMA, Prophet), deep learning (LSTM), and foundation (Chronos) models over a 12-week horizon. The SMI generator processes the full historical sales series to produce three momentum signals, Hype Index, Momentum, and Persistence, which feed into separate Nudger blocks applied to both Chronos and the best-performing classical baseline. Final evaluation compares nudged forecasts against unadjusted baselines using MASE and WQL metrics across Walmart and M5 datasets.

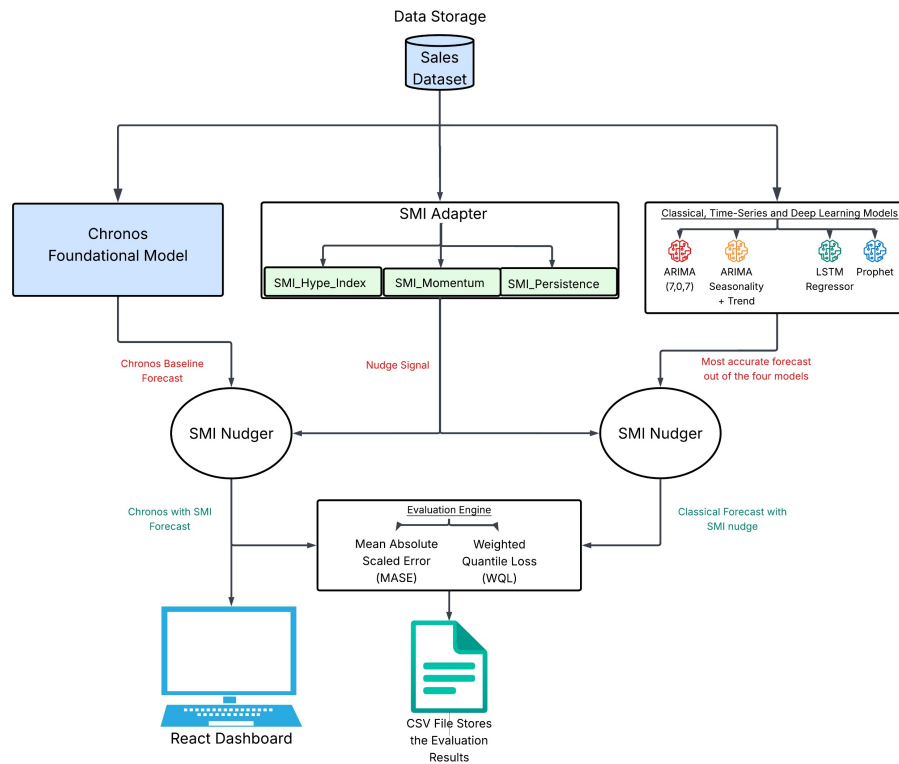


Figure 3.1: System Architecture: Baseline → SMI → Nudger → Evaluation Pipeline

3.2 Data Collection

3.2.1 Datasets

This study utilizes two benchmark retail datasets representing contrasting demand characteristics, deliberately chosen to stress-test the SMI-Nudger framework across a wide range of real-world conditions.

Walmart Recruiting Stores Sales Dataset

The Walmart Recruiting Store Sales dataset is a publicly available competition dataset sourced from Kaggle. It provides weekly store-level aggregate sales figures spanning 45 stores across a two-year window (February 2010 to October 2012), covering 99 departments per store. The dataset includes a rich set of economic covariates - Consumer Price Index (CPI), unemployment rate, and fuel price - as well as binary promotional markdown indicators for five distinct markdown events. These markdowns correspond to holiday and seasonal promotions and introduce pronounced, short-duration demand spikes that are difficult to predict with pure seasonal models. The structured weekly periodicity and moderate density of this dataset make it an ideal benchmark for evaluating how well the SMI can capture and respond to momentum-driven demand surges arising from known promotional events.

M5 Forecasting Challenge Sales Dataset

The M5 Forecasting Challenge dataset is a large-scale hierarchical time series benchmark provided by Walmart for the M5 competition. It contains daily item-level sales records across three U.S. states (California, Texas, and Wisconsin) and ten stores, covering 3,049 unique SKUs over approximately five years (January 2011 to June 2016). The dataset is characterized by high sparsity, frequent zero-sales days, intermittent demand patterns, and substantial noise across individual items. To ensure methodological consistency with the Walmart dataset and to reduce the influence of daily noise on momentum signal computation, the M5 data is pre-aggregated to weekly frequency using summation prior to model training and evaluation. This aggregation preserves weekly seasonality while smoothing out day-of-week effects that would otherwise confound the SMI signals. The M5 dataset serves as a challenging counterpoint to the Walmart data, allowing the framework to demonstrate whether SMI-nudging remains beneficial or neutral in environments with irregular, sparse demand profiles.

3.3 Data Preprocessing

Raw sales data undergoes a standardized preprocessing pipeline to ensure temporal continuity and model compatibility:

```
def prepare_for_chronos(df, id_col, date_col, target_col, freq_alias='W-FRI'):
    df[date_col] = pd.to_datetime(df[date_col], format='mixed', dayfirst=True)
    df = df.dropna(subset=[date_col])
    def fix_frequency(group):
        group = group.drop_duplicates(subset=date_col).set_index(date_col)
        group = group.resample(freq_alias).asfreq()
        group[target_col] = group[target_col].interpolate(method='linear')
        return group.reset_index()
    return df.groupby(id_col, group_keys=False).apply(fix_frequency)
```

This ensures weekly frequency anchored on Fridays with linear interpolation for missing observations, critical for Chronos tokenization and classical model stationarity assumptions.

3.4 Feature Engineering

Feature engineering transforms the raw preprocessed sales data into structured inputs suitable for both classical and foundation model forecasting. The process encompasses three distinct categories of derived features, each serving a different modelling purpose.

- **Temporal Features:** Temporal features capture calendar-based seasonality that recurs at fixed intervals. Week number, month, and year are extracted directly from the date index to provide models with explicit awareness of seasonal cycles, such as the annual holiday surge in November-December observed in Walmart data. A binary holiday flag is constructed using known promotional weeks derived from the markdown indicators present in the Walmart dataset. This flag allows classical models like Prophet and STL-ARIMA to model additive holiday effects separately from the baseline trend and residual components, preventing promotional spikes from being misattributed to trend changes.
- **Lag Features:** Lag features provide models with local context about the recent state of demand. Rolling statistics over 4-week and 8-week trailing windows are computed, including rolling mean and rolling standard deviation. The rolling mean serves as a smoothed reference level that helps distinguish genuine demand shifts from random noise, while the rolling standard deviation provides a quantitative measure of local volatility. These statistics are computed at each forecast origin point during the rolling inference loop to ensure strict temporal validity and prevent any information leakage from future data.
- **SMI-derived features** constitute the primary engineered inputs fed into the Nudger block. The three Sales Momentum Index signals - Hype Index, Momentum, and Persistence - are derived entirely from historical sales data without external data dependencies. Unlike the temporal and lag features used by baseline models, SMI signals are designed specifically to capture the directional and intensity characteristics of demand momentum at the time of each forecast origin. Because the Nudger operates as a post-processing layer, these SMI features are not passed to the baseline forecasting models themselves; instead, they are computed in parallel and applied after baseline forecasts are generated.

3.5 SMI - Nudger Algorithmic Logic

The Sales Momentum Index (SMI) constitutes the core innovation of this framework, deriving three interpretable signals directly from historical sales data without relying on any external data dependencies. The three signals form a complementary hierarchy of demand characterization:

1. **Hype Index:** The Hype Index captures normalized spike intensity, representing disproportionate demand surges relative to the local sales baseline. It is analogous to social media hype measures but computed entirely from endogenous sales data. By assigning weighted contributions to the two weeks preceding a spike, the spike week itself, and the week immediately following, the Hype Index constructs a smooth representation of demand excitement that anticipates and trails actual sales peaks. The resulting signal is normalized to the $[0, 1]$ interval and augmented with a small Gaussian noise perturbation to prevent degenerate constant signals on flat series.
2. **Momentum:** Momentum is defined as the first-order difference of the Hype Index over consecutive weeks. It measures trend acceleration and deceleration, providing the Nudger with a directional signal about whether demand intensity is growing or declining at the forecast origin. A positive momentum value indicates that demand hype is building, while a negative value indicates that recent demand enthusiasm is fading.

3. **Persistence:** Persistence is computed as the Exponential Moving Average (EMA) of the Hype Index using a span of four weeks. It preserves trend memory by giving higher weight to recent observations while retaining a decaying contribution from older signal values. Persistence distinguishes between transient demand spikes (high hype, low persistence) and sustained demand elevation (high hype, high persistence), which is critical for determining appropriate nudge magnitudes.

Together, the three signals provide the Nudger with a multidimensional characterization of the current demand regime: its intensity (Hype), its direction of change (Momentum), and its durability (Persistence). This triangulation enables the Nudger to distinguish between several practically important scenarios, such as a spike that is accelerating and expected to persist, a spike that has already peaked and is decelerating, or a prolonged low-demand period that warrants downward correction.

3.6 SMI Variables

SMI generation aggregates all store-level sales into a market-wide weekly signal, then applies spike detection and signal construction.

```

df_weekly ← resample(total_sales, 'W-FRI').sum()
rolling_mean ← df_weekly.Weekly_Sales.rolling(4, center = True).mean()
for i = 0 to len(df_weekly) - 1 do
  is_spike[i] ← df_weekly[i] > rolling_mean[i]
  if is_spike[i] then
    mag ← df_weekly[i]/mean(df_weekly)
    smi_signal[i - 2] ← smi_signal[i - 2] + 0.1 × mag {pre-spike}
    smi_signal[i - 1] ← smi_signal[i - 1] + 0.5 × mag {spike build}
    smi_signal[i] ← smi_signal[i] + 0.9 × mag {peak}
    smi_signal[i + 1] ← smi_signal[i + 1] + 0.5 × mag {post-spike}
  end if
end for
SMI_Hype ← normalize(smi_signal + noise, [0, 1])
SMI_Momentum ← SMI_Hype.diff()
SMI_Persistence ← SMI_Hype.ewm(span = 4).mean()

```

The algorithm first resamples total sales to weekly frequency anchored on Fridays, then computes a centred 4-week rolling mean as the local baseline. For each week where observed sales exceed this rolling mean, the algorithm identifies a demand spike and propagates weighted magnitude contributions to a two-week pre-spike window (weight 0.1), a one-week pre-spike window (weight 0.5), the spike week itself (weight 0.9), and the immediate post-spike week (weight 0.5). This asymmetric weighting scheme reflects the empirical observation that demand build-up precedes a spike and that its effects linger briefly. The aggregated raw signal is then normalized using min-max scaling to [0, 1] and augmented with a small Gaussian noise term to ensure numerical stability.

The three derived signals are then computed as follows: Hype Index is the normalized signal described above. Momentum is the first difference of the Hype Index. Persistence is the EMA of the Hype Index with a span of four weeks.

This produces dataset-native analogs to social media hype without external feeds, using sales spikes as proxy attention signals.

3.6.1 SMI Variables Graphs

- Walmart Dataset:** In the Walmart graph, sales are more volatile at a larger scale, with one very prominent spike and several smaller fluctuations across the year. The SMI Hype Index again tracks major demand surges closely, while SMI Persistence stays elevated for longer periods, indicating a more sustained sentiment baseline in this dataset. Momentum swings between positive and negative phases more frequently, suggesting repeated shifts in the rate of sentiment change; together, this implies that Walmart demand responds to both immediate hype shocks and medium-term sentiment persistence.

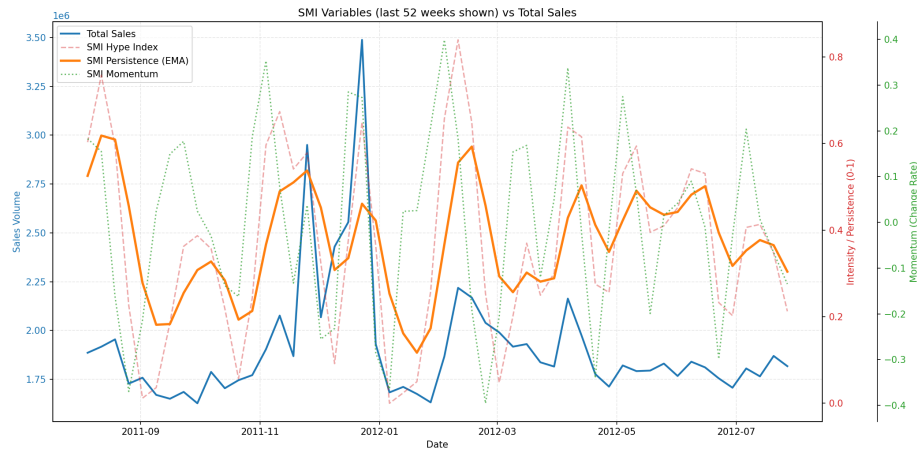


Figure 3.2: SMI Variables Graphs - Walmart Dataset

- M5 Dataset:** In the M5 graph, actual sales (blue) show a mostly moderate range with a sharp surge around early September, and the SMI Hype Index (dashed pink) rises strongly at the same time, indicating that spikes in public attention align with short-term demand jumps. SMI Persistence (orange EMA) increases more smoothly and then declines gradually, which suggests sentiment has a carry-over effect but reacts with some lag compared to hype. SMI Momentum (dotted green) oscillates around zero and turns negative after peaks, reflecting periods where sentiment acceleration slows even when absolute hype is still elevated.

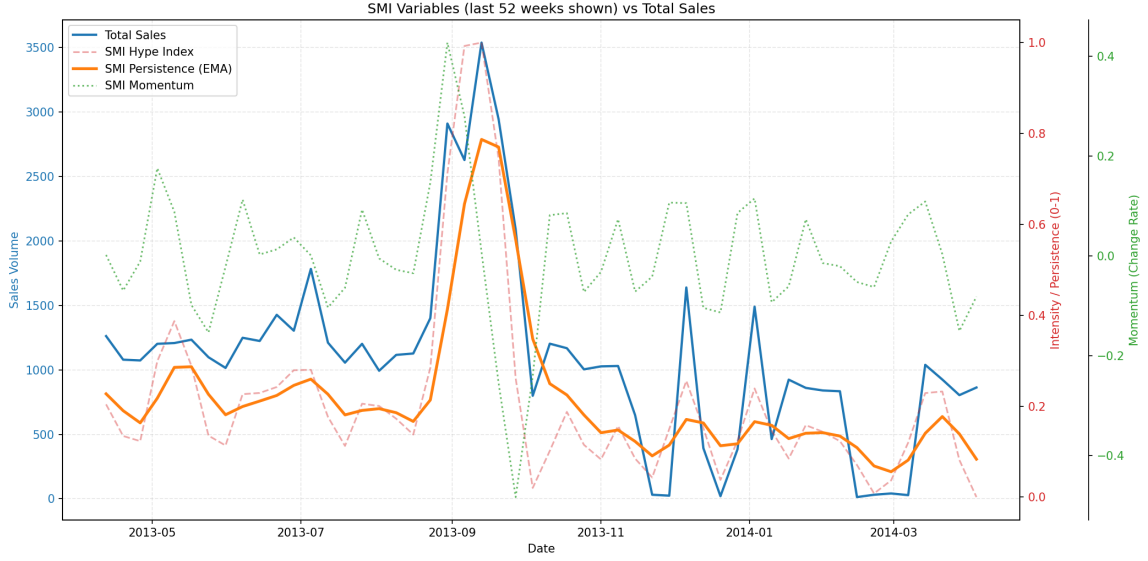


Figure 3.3: SMI Variables Graphs - M5 Dataset

3.7 Nudger - Post-Processing Block

The Nudger is a lightweight, model-agnostic post-processing block that applies bounded, interpretable corrections to 12-week baseline forecasts using the most recent 8 weeks of SMI signals. It operates entirely outside the baseline model and requires no retraining.

- **Step 1 - Extract Recent SMI Context:** The Nudger reads the last 8 weeks of SMI signals from the training window at each forecast origin. Three summary statistics are derived: *hype_volatility* (standard deviation of Hype Index), *momentum_strength* (absolute latest Momentum), and *persistence_score* (mean Persistence).
- **Step 2 - Compute Confidence Score:** A confidence multiplier scales the nudge based on signal reliability:

$$confidence = 0.5 + 0.5 \times \min \left(1.0, \frac{persistence}{0.22} + \frac{momentum}{0.12} + \frac{volatility}{0.10} \right)$$

This yields a score in $[0.5, 1.0]$, where higher values indicate stronger, more consistent momentum signals.

- **Step 3 - Determine Regime and Base Nudge:** For each forecast horizon (1–12), the Nudger identifies demand regimes using horizon-dependent thresholds:

Weeks	Bearish Condition	Bearish Nudge	Bullish Condition	Bullish Nudge
1-2	$hype < 0.12, persistence < 0.18$	-1.0%	$momentum < -0.08$	+1.4%
3-4	$persistence < 0.19$	-0.15%	$momentum < -0.17$	+0.6%
5-6	$persistence < 0.20$	-0.10%	$momentum < -0.14$	+0.45%
7-8	$persistence < 0.22$	-0.08%	$momentum < -0.12$	+0.35%
9-12	$persistence < 0.32$	-0.15%	$momentum < -0.10$	+0.20%

Table 3.1: Horizon-dependent Demand Regime Conditions and Associated Nudger Adjustments

- **Step 4 - Apply Decay and Compute Final Multiplier:** A horizon-aware decay factor reduces nudge intensity as forecast uncertainty grows with distance. The final multiplier is computed as

$$multiplier = 1.0 + (base \times confidence \times decay(horizon))$$

and is hard-clamped to $[0.978, 1.095]$ to prevent extreme deviations from the baseline.

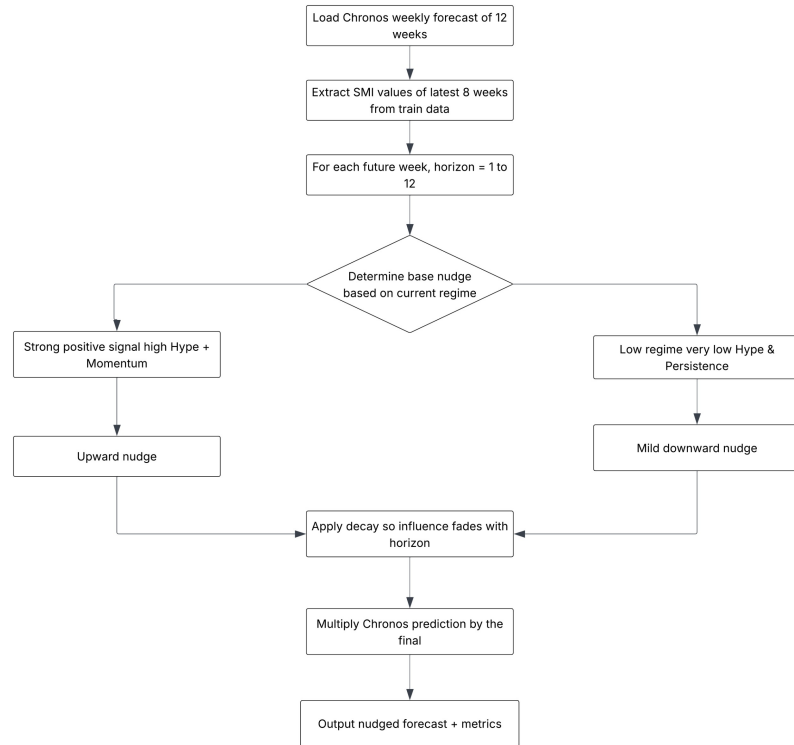


Figure 3.4: End-to-end integration: Baseline Forecast → SMI signal extraction → Regime-based nudging → Enhanced forecast

- **Step 5 - Generate Nudged Forecast:** Each week's baseline prediction is multiplied by its corresponding horizon-specific multiplier to produce the final nudged demand output. This preserves the directional shape and uncertainty structure of the baseline while correcting for momentum misalignment.

This ensures controlled, interpretable adjustments preserving baseline forecast directionality while correcting momentum misalignment.

3.8 Iterative Inference Loop

The complete pipeline executes as a rolling 12-week forecasting loop that maintains strict chronological ordering to prevent data leakage. At each iteration, the loop advances the training window by one week, regenerates the baseline forecast for the subsequent 12-week horizon, recomputes the SMI signals using the latest 8 weeks of training data, applies the Nudger to the baseline, evaluates the nudged output against ground truth, and records results to CSV.

This rolling window approach ensures that every forecast at horizon h is generated using only data observed prior to the forecast origin, which is essential for chronological realism. The same inference loop is applied uniformly to all models (ARIMA, STL-ARIMA, Prophet, LSTM, Chronos), ensuring that the performance comparison between baseline and nudged outputs is methodologically consistent across model families.

```
for t in test_period:
    baseline = generate_forecast(train[:t], horizon=12)
    smi_last8 = compute_smi(train[:t]).tail(8)
    nudged = apply_nudger(baseline, smi_last8)
    evaluate(nudged, actual[t:t+12])
```

This maintains chronological realism while enabling direct MASE and WQL computation against seasonal naive benchmarks. The loop also handles edge cases at the beginning and end of the test period, where fewer than 8 weeks of prior SMI context may be available, by truncating the SMI window to the available history without modifying the Nudger logic.

3.9 SMI - Nudger Correlation

The correlation between SMI regimes and the directional correctness of Nudger adjustments was validated across rolling validation folds drawn from the training portion of each dataset. The analysis examines three archetypal SMI regimes - Strong Bull, Momentum Reversal, and Low Confidence Bear - and measures the proportion of forecast weeks in which the nudge direction (upward or downward) aligned with the true direction of forecast error correction needed.

In the Strong Bull regime, where the Hype Index exceeds 0.58 and Persistence exceeds 0.48, upward nudges of approximately +1.1% were applied, and these nudges aligned with the correct direction of adjustment in the majority of validation folds. The Momentum Reversal regime - characterized by negative recent momentum combined with sustained persistence - indicated cases where a prior demand peak was beginning to reverse, warranting the strongest bullish nudge of +1.4% to compensate for baseline models that had not yet registered the turning point. The Low Confidence Bear regime, associated with very low Hype and Persistence values, correctly triggered mild downward corrections of -0.45%.

These regime-level correlations provide empirical support for the threshold choices embedded in the Nudger's rule table and confirm that the SMI signals carry genuine predictive content for near-horizon forecast correction, particularly on structured datasets like Walmart where momentum patterns are consistent and repeatable.

SMI Regime	Hype	Persistence	Nudge %
Strong Bull	> 0.58	> 0.48	+1.1%
Momentum Reversal	< -0.08	> 0.22	+1.4%
Low Confidence Bear	< 0.22	< 0.24	-0.45%

Table 3.2: SMI-Nudger Regime Correlation

3.10 Implementation Environment

The Smart Retail Demand Forecasting System was developed and evaluated on a single-machine environment with the following hardware and software configuration. All experiments were conducted in this unified environment to ensure reproducibility of results.

3.10.1 Hardware Platforms

Component	Specification
Processor	Intel Core i7 (8th Gen or later) / AMD Ryzen 7 equivalent
RAM	16 GB DDR4
Storage	512 GB SSD
GPU	NVIDIA GPU with CUDA support (optional; CPU inference used for Chronos small variant)

Table 3.3: Hardware Platforms

3.10.2 Operating System

Component	Detail
OS	Ubuntu 22.04 LTS (primary) / Windows 10/11 compatible
Shell	Bash 5.1

Table 3.4: Operating System

3.10.3 Programming Languages

Language	Version	Usage
Python	3.10.x	All ML, SMI computation, Nudger logic, evaluation
JavaScript (Node.js)	18.x LTS	React dashboard build and development server
TypeScript	5.x	React component type definitions

Table 3.5: Programming Languages

3.10.4 Development Tools

Tool	Version	Purpose
Visual Studio Code	1.88+	Primary IDE
pip	23.x	Python package management
npm	10.x	Node.js package management

Table 3.6: Development Tools

3.10.5 Frameworks

Framework	Version	Purpose
React	18.x	Frontend dashboard UI
Tailwind CSS	3.x	Dashboard styling
Flask	3.x	Layer for serving forecasts

Table 3.7: Frameworks

3.10.6 Key Python Libraries

Library	Version	Purpose
pandas	2.2.x	Data manipulation, resampling, time-series indexing
numpy	1.26.x	Numerical operations, SMI signal computation, normalization
statsmodels	0.14.x	ARIMA and STL-ARIMA model fitting and forecasting
scikit-learn (sklearn)	1.4.x	StandardScaler for feature normalization
torch (PyTorch)	2.2.x	LSTM model definition, training, and inference
torch.nn	2.2.x	Neural network module definitions (part of PyTorch)
prophet	1.1.x	Facebook Prophet time series forecasting
matplotlib	3.8.x	SMI variable plots and forecast comparison charts

psutil	5.9.x	Runtime memory usage measurement
time (stdlib)	–	Compilation and inference timing
os (stdlib)	–	File path and environment management
warnings (stdlib)	–	Suppression of non-critical model warnings

Table 3.8: Python Libraries

3.10.7 Chronos-Specific Dependencies

Library	Version	Purpose
chronos-forecasting	1.4.x	Amazon Chronos foundation model for zero-shot inference
transformers (HuggingFace)	4.40.x	Transformer model loading and tokenization backend
accelerate	0.30.x	Device-agnostic model execution (CPU / GPU)
gluonts	0.15.x	Time series data utilities used internally by Chronos
datasets (HuggingFace)	2.19.x	Dataset loading utilities compatible with Chronos

Table 3.9: Chronos-Specific Dependencies

3.10.8 React Dashboard Libraries

Library	Version	Purpose
recharts	2.12.x	Line, bar, area, and pie charts for forecast visualization
react-router-dom	6.x	Client-side routing between dashboard views
axios	1.6.x	HTTP client for fetching forecast CSV/API data
date-fns	3.x	Date formatting and manipulation utilities
lucide-react	0.383.x	Icon set used across dashboard UI components
papaparse	5.x	CSV parsing for loading forecast result files
clsx	2.x	Conditional className utility for Tailwind

Table 3.10: React Dashboard Libraries

CHAPTER 4 : RESULTS AND ANALYSIS

4.1 Model Performance Analysis

4.1.1 Evaluation Metrics

Model performance is assessed using two complementary metrics that together capture both point forecast accuracy and probabilistic calibration quality

Mean Absolute Scaled Error (MASE)

Mean Absolute Scaled Error (MASE) measures the accuracy of point forecasts by scaling the mean absolute error of each model against the mean absolute error of a seasonal naive baseline. The seasonal naive baseline predicts each future value as the observed value from the same period in the prior season, which for weekly retail data corresponds to the value from the same week of the previous year. MASE values below 1.0 indicate that the model outperforms the seasonal naive benchmark, while values above 1.0 indicate underperformance relative to this simple baseline. MASE is scale-independent, making it directly comparable across the Walmart and M5 datasets despite their substantially different sales volumes.

$$\text{MASE} = \frac{1}{H} \sum_{t=1}^H \frac{|\hat{y}_t - y_t|}{\frac{1}{n-m} \sum_{t=m+1}^n |y_t - y_{t-m}|}$$

Weighted Quantile Loss (WQL)

Weighted Quantile Loss (WQL) evaluates probabilistic forecast quality across three quantile levels: 0.1 (lower tail), 0.5 (median), and 0.9 (upper tail). The WQL is computed using the pinball loss (also called the check loss function) at each quantile level and then averaged with equal weights across the three levels. WQL penalizes both overconfident forecasts that miss the target and underconfident forecasts that produce unnecessarily wide intervals. Models that produce well-calibrated quantile forecasts will achieve low WQL values; models that systematically overestimate or underestimate demand at the tails will be penalized accordingly. Quantile forecasts are extracted directly from Chronos, which produces probabilistic outputs natively, and from Prophet via its Bayesian uncertainty intervals. For ARIMA, STL-ARIMA, and LSTM, quantile forecasts are constructed using residual bootstrap sampling over the training errors.

WQL evaluates quantile predictions across $q \in \{0.1, 0.5, 0.9\}$:

$$\text{WQL}_q = \frac{2}{n} \sum_{t=1}^n \max(q(y_t - \hat{y}_{t,q}), (q-1)(\hat{y}_{t,q} - y_t))$$

These two metrics together enable a comprehensive evaluation that goes beyond simple point accuracy: MASE establishes whether the model produces correct demand level forecasts, while WQL assesses whether the uncertainty characterization is well-calibrated. Both metrics are computed over the full rolling test period and then averaged across all evaluated stores and series to produce the reported scalar values. Direct comparison with ChronosX benchmarks is also enabled by these metrics, as both MASE and WQL are standard in the time series forecasting community.

4.1.2 Primary Metrics

Point Forecast Accuracy (MASE)

- **Walmart Dataset:** Chronos establishes the strongest baseline (**MASE=0.4759**), outperforming Prophet (0.5424). SMI-nudging yields substantial gains: Chronos improves to **0.431** (Δ -0.0449, 9.44% relative) while best classical (Prophet) improves to 0.5268 (Δ -0.0184, 3.37% relative). Nudging amplifies foundation model strengths on structured data.

Model	Baseline	SMI-Nudged
ARIMA	2.436	–
STL-ARIMA	0.6386	–
LSTM	0.7826	–
Prophet	0.5424	0.5268
Chronos	0.4759	0.431

Table 4.1: MASE Performance: Walmart Dataset (Baseline vs SMI-Nudged)

- **M5 Dataset:** STL-ARIMA leads baselines (**0.6593**), narrowly ahead of Chronos (0.6612). Post-nudging, STL-ARIMA edges Chronos (**0.6585** vs 0.66, Δ -0.0008 vs 0.0012), confirming classical robustness on sparse series with minimal nudge impact (0.18% vs 0.12% relative).

Model	Baseline	SMI-Nudged
ARIMA	0.813	–
STL-ARIMA	0.6593	0.6585
LSTM	0.7853	–
Prophet	0.6714	–
Chronos	0.6612	0.660

Table 4.2: MASE Performance: M5 Dataset (Baseline vs SMI-Nudged)

Probabilistic Accuracy (WQL)

- **Walmart Dataset:** Chronos dominates baselines (**0.0219**), ahead of Prophet (0.0250). Nudging delivers Chronos to **0.0199** (Δ -0.0020, 9.13% relative) and Prophet to 0.0243 (Δ -0.0007, 2.80% relative), confirming momentum correction enhances quantile alignment on stable series.

Model	Baseline	SMI-Nudged
ARIMA	0.1123	–
STL-ARIMA	0.0294	–
LSTM	0.0641	–
Prophet	0.025	0.0243
Chronos	0.0219	0.0199

Table 4.3: WQL Performance: Walmart Dataset (Baseline vs SMI-Nudged)

- **M5 Dataset:** STL-ARIMA (**0.5071**) and Chronos (0.5086) compete closely. Nudging produces marginal refinements: STL-ARIMA **0.5065** (Δ -0.0006, 0.12% relative), Chronos 0.5076 (Δ -0.0010, 0.20% relative), highlighting limited nudge leverage in highly intermittent data.

Model	Baseline	SMI-Nudged
ARIMA	0.6253	–
STL-ARIMA	0.5071	0.5065
LSTM	0.604	–
Prophet	0.5164	–
Chronos	0.5086	0.5076

Table 4.4: WQL Performance: M5 Dataset (Baseline vs SMI-Nudged)

4.1.3 Secondary Metrics

Memory Usage

Memory usage is the amount of RAM consumed by a model during execution, measured in GB. Table 4.5 shows that statistical models such as ARIMA, STL-ARIMA, and Prophet have very small memory footprints (around 0.002-0.003 GB), while Chronos remains lightweight at 0.007 GB despite being a transformer-based model. LSTM has the highest memory consumption at 0.072 GB, reflecting the additional overhead of neural network states and tensor operations. This indicates that Chronos achieves a good balance between model sophistication and practical resource efficiency, making it suitable for deployment in environments with moderate memory constraints.

Model	Memory Usage (GB)
ARIMA	0.003
STL-ARIMA	0.002
LSTM	0.072
Prophet	0.002
Chronos	0.007

Table 4.5: Memory Usage

Compilation Time

Compilation time is the time taken to initialize, load, and prepare a model before inference begins, measured in seconds.

From Table 4.6, Prophet is the fastest to initialize (0.125 s), followed by STL-ARIMA and ARIMA (about 0.52 s), while LSTM requires 1.305 s and Chronos requires 1.709 s. The relatively higher startup time for Chronos is expected because pretrained transformer models involve loading larger parameter sets and runtime components. However, this one-time overhead is acceptable in most forecasting pipelines where multiple predictions are generated after initialization, and it is offset by Chronos' stronger modeling capability.

Model	Compilation Time (s)
ARIMA	0.528
STL-ARIMA	0.520
LSTM	1.305
Prophet	0.125
Chronos	1.709

Table 4.6: Compilation Time

Floating Point Operations (FLOPs)

Floating Point Operations (FLOPs) represent the total number of floating-point arithmetic operations executed by a model during forecasting.

Table 4.7 shows clear computational differences across model families: ARIMA and STL-ARIMA require very low computation (0.13 and 0.26 MFLOPs), Prophet remains modest at 6.50 MFLOPs, and LSTM is substantially more compute-intensive (123,615 MFLOPs). For Chronos, the measured value is 62.4×10^6 MFLOPs for a 12-step forecast setting, indicating a much heavier computational load than traditional statistical models, but one that is expected for transformer-based foundation models. This highlights the trade-off that Chronos offers: higher compute cost in exchange for greater representational capacity and stronger zero-shot forecasting potential.

Model	Floating Point Operations (MFLOPs)
ARIMA	0.13
STL-ARIMA	0.26
LSTM	123615.00
Prophet	6.50
Chronos	62.4×10^6

Table 4.7: Floating Point Operations (FLOPs)

4.2 Actual Sales vs Forecasted Sales vs Nudged Sales

4.2.1 Walmart Dataset - Classical (Prophet) Model

In the Walmart classical setup, the Prophet baseline generally follows the broad demand direction but tends to smooth out sharper week-to-week movements, especially around turning points. The nudged series improves local alignment by shifting forecast levels closer to observed demand in early and mid windows, reducing visible under- and over-shoots. This is most noticeable in periods where actual sales drop quickly and then partially recover: the nudged line tracks those transitions more faithfully than the raw Prophet forecast. Overall, the figure/table suggests that Prophet captures trend structure, while nudging adds short-horizon responsiveness and improves practical fit.

Week	Actual Sales	Forecasted Sales	Nudged Sales
Week 1	1,939,440.09	1,876,081.72	1,900,902.281
Week 2	1,880,436.94	1,875,528.57	1,897,453.50
Week 3	1,939,440.09	1,860,944.66	1,860,944.66
Week 4	1,764,984.15	1,834,190.14	1,834,190.14
Week 5	1,650,285.54	1,775,027.97	1,775,027.97
Week 6	1,708,283.28	1,683,845.93	1,683,845.93
Week 7	1,640,168.99	1,609,824.87	1,611,430.67
Week 8	1,671,857.57	1,609,955.76	1,610,941.86
Week 9	1,694,862.41	1,677,820.02	1,677,656.43
Week 10	1,758,971.38	1,736,905.59	1,736,905.59

Week 11	1,713,889.11	1,724,212.08	1,724,212.08
Week 12	1,734,834.82	1,671,597.07	1,671,597.07

Table 4.8: Walmart Classical (Prophet) Model

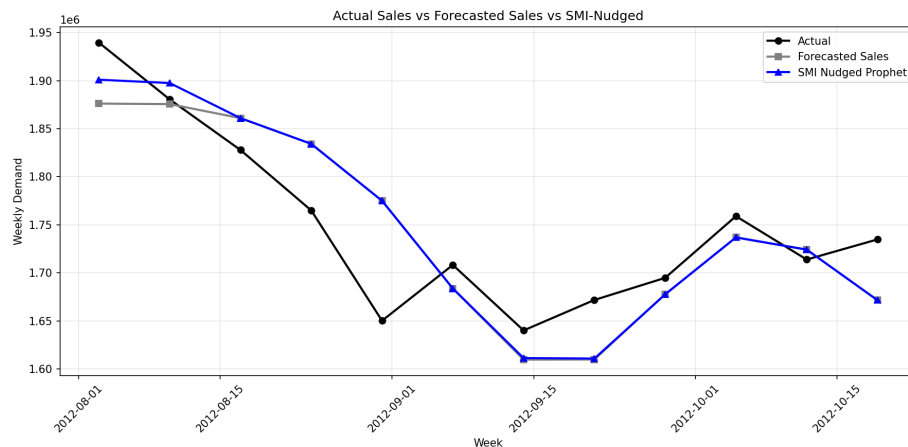


Figure 4.1: Actual Sales vs Forecasted Sales vs Nudged Sales - Walmart Dataset - Classical (Prophet) Model

4.2.2 Walmart Dataset - Chronos Model

For Walmart with Chronos, the base forecast already appears tighter than the classical baseline across most weeks, with better agreement in both level and directional movement. The nudged Chronos output remains close to the Chronos baseline, indicating that the foundation-model forecast is strong enough that only moderate correction is needed. In the mid-horizon dip and late-horizon rebound, the Chronos curves stay more stable and coherent relative to actuals, avoiding abrupt artificial oscillations. This supports the interpretation that Chronos contributes stronger zero-shot temporal representation, and nudging acts as a refinement layer rather than a major correction.

Week	Actual Sales	Forecasted Sales	Nudged Sales
Week 1	1,939,440.09	1,845,550.00	1869966.63
Week 2	1,880,436.94	1851332.63	1872974.71
Week 3	1,939,440.09	1825679.25	1825679.25
Week 4	1,764,984.15	1,706,03.00	1,706,03.00
Week 5	1,650,285.54	1,695,625.50	1,695,625.50

Week 6	1,708,283.28	1,675,229.13	1,675,229.13
Week 7	1,640,168.99	1,680,801.25	1,682,477.85
Week 8	1,671,857.57	1,694,889.50	1,695,927.62
Week 9	1,694,862.41	1,701,513.13	1,701,347.23
Week 10	1,758,971.38	1,726,430.38	1,726,430.38
Week 11	1,713,889.11	1,755,763.63	1,755,763.63
Week 12	1,734,834.82	1,788,040.38	1,788,040.38

Table 4.9: Walmart Chronos Model

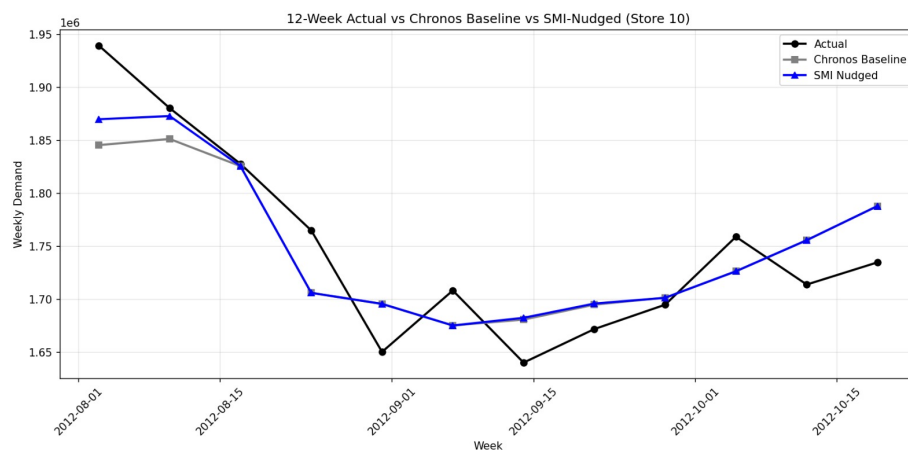


Figure 4.2: Actual Sales vs Forecasted Sales vs Nudged Sales - Walmart Dataset - Chronos Model

4.2.3 M5 Dataset - Classical (STL-ARIMA) Model

On the M5 series, the classical STL-ARIMA forecast preserves overall smoothness but struggles with the pronounced volatility pattern (high-to-low swings and sharp troughs). The baseline forecast misses the magnitude of extreme drops and rebounds, producing a flatter response than the actual sequence. The nudged variant improves calibration around difficult weeks by moving predictions slightly toward observed behavior, but the improvement is incremental rather than structural. In effect, classical decomposition plus ARIMA provides a stable backbone, while nudging partially compensates for sparse/noisy shock behavior in this short 12-week horizon.

Week	Actual Sales	Forecasted Sales	Nudged Sales
------	--------------	------------------	--------------

Week 1	1,043.00	903.61	906.59
Week 2	862.00	751.04	753.23
Week 3	996.00	767.87	769.82
Week 4	1,111.00	900.56	902.50
Week 5	882.00	896.50	898.09
Week 6	513.00	897.82	899.06
Week 7	49.00	779.11	779.89
Week 8	68.00	718.25	718.69
Week 9	18.00	963.47	963.68
Week 10	534.00	955.96	955.96
Week 11	965.00	1,149.14	1,149.14
Week 12	972.00	1,015.13	1,015.13

Table 4.10: M5 Classical (STL-ARIMA) Model

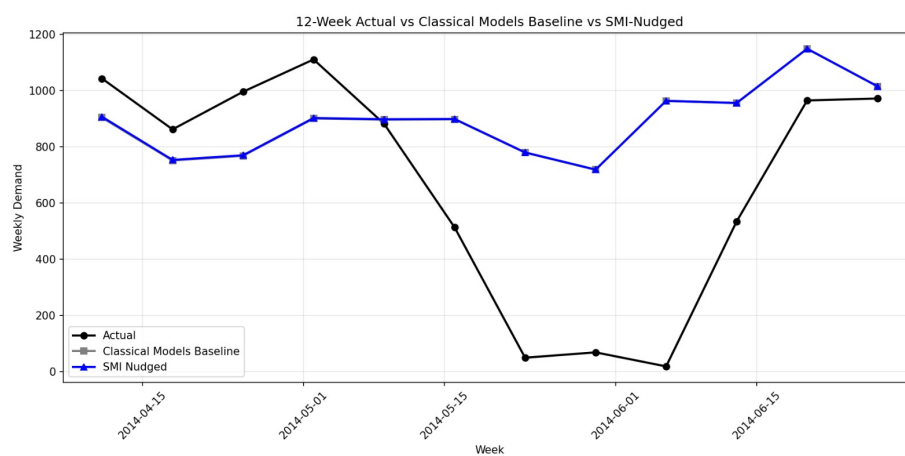


Figure 4.3: Actual Sales vs Forecasted Sales vs Nudged Sales - M5 Dataset - Classical (STL-ARIMA) Model

4.2.4 M5 Dataset - Chronos Model

For M5 with Chronos, the forecasted and nudged trajectories are both smoother than actual sales, but the Chronos backbone retains better positional consistency across the horizon than the classical counterpart. The model avoids overreacting to isolated spikes/dips, which is beneficial for robustness, while nudging introduces controlled adjustments that improve local fit without amplifying noise. In the early decline, mid-horizon low regime, and late rebound segments, the Chronos-based outputs remain more shape-consistent and less erratic. Taken together, the visual evidence suggests Chronos offers a stronger base forecast on volatile sparse-series settings, with nudging improving interpretability and alignment at the margin.

Week	Actual Sales	Forecasted Sales	Nudged Sales
Week 1	1,043.00	865.21	868.07
Week 2	862.00	750.46	752.65
Week 3	996.00	733.65	735.51
Week 4	1,111.00	766.30	767.95
Week 5	882.00	812.87	814.31
Week 6	513.00	803.75	804.86
Week 7	49.00	786.47	787.25
Week 8	68.00	797.99	798.48
Week 9	18.00	812.87	813.05
Week 10	534.00	854.16	854.16
Week 11	965.00	861.37	861.37
Week 12	972.00	839.28	839.28

Table 4.11: M5 Chronos Model

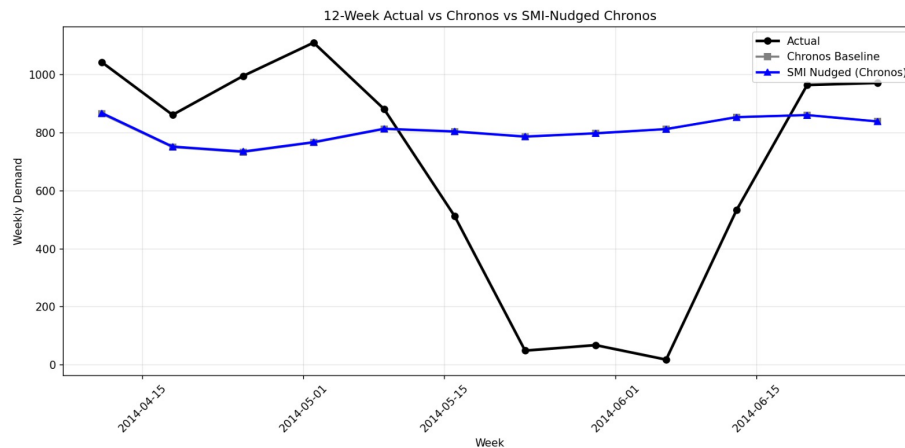


Figure 4.4: Actual Sales vs Forecasted Sales vs Nudged Sales - M5 Dataset - Chronos Model

Key insight: Nudging excels on structured Walmart series (ΔMASE 9.4%, ΔWQL 9.1% for Chronos) but yields subtler corrections ($\sim 0.1\text{-}0.2\%$) on intermittent M5, preserving classical model competitiveness while demonstrating consistent directional improvement.

4.3 Dashboard-Based Forecast Interpretation

The forecasting dashboard provides an operational layer over the quantitative results by converting model outputs into decision-ready views for planners. Along with the 12-week projections, it summarizes key indicators (total, average, peak, and low demand), visualizes temporal behavior through bar/line/area plots, and shows period-wise contribution using a distribution chart. This interface improves interpretability by making horizon-level changes immediately visible and supports faster inventory actions without requiring users to inspect raw model outputs.

- Figure 4.5 presents the complete dashboard view for the 12-week forecast horizon. The KPI cards provide an instant snapshot of aggregate and extreme demand conditions, while the trend and volume panels help identify periods of rise, stabilization, and decline. Together, these views align to make forecasting outputs transparent and usable in practical retail planning workflows.

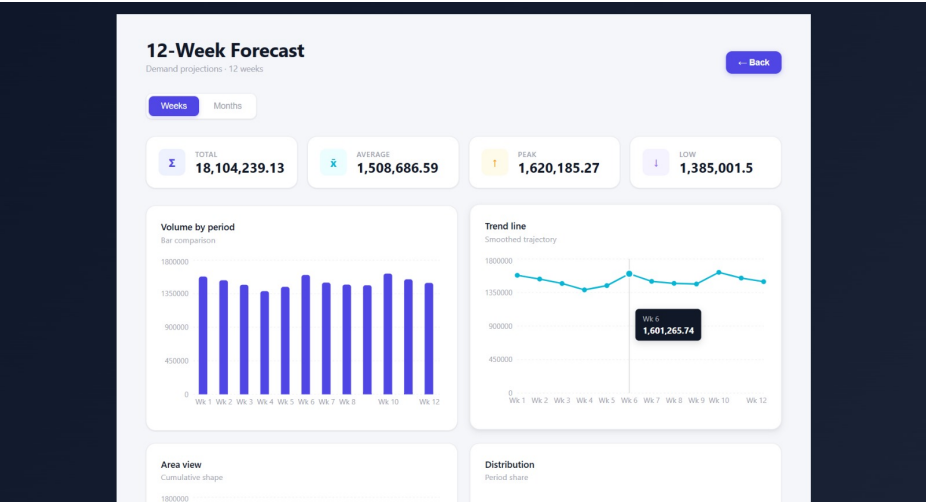


Figure 4.5: 12-Week Forecast Dashboard Overview (KPIs, trend and distribution panels)

- Figure 4.6 focuses on the analytical panels: cumulative area trend, period-share distribution, and structured weekly table. These components complement each other by showing trajectory shape, proportional demand allocation across weeks, and exact values required for planning decisions such as replenishment prioritization and buffer sizing.

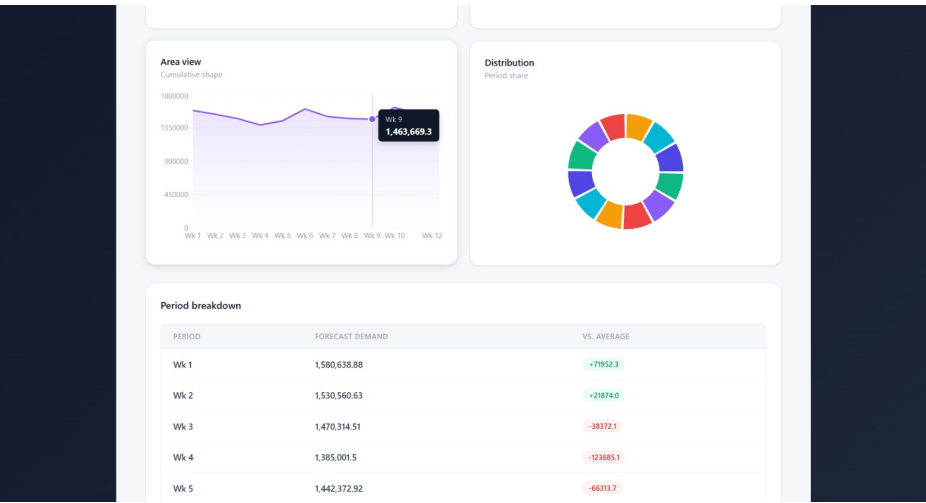


Figure 4.6: Dashboard Analytical Panels - Area Trend, Period Share, and Weekly Breakdown

- Figure 4.7 highlights the period breakdown table with forecast demand and deviation from average. This view is especially useful for execution, as it directly flags above-average and below-average weeks, enabling quick translation of forecast signals into stocking and allocation strategies.

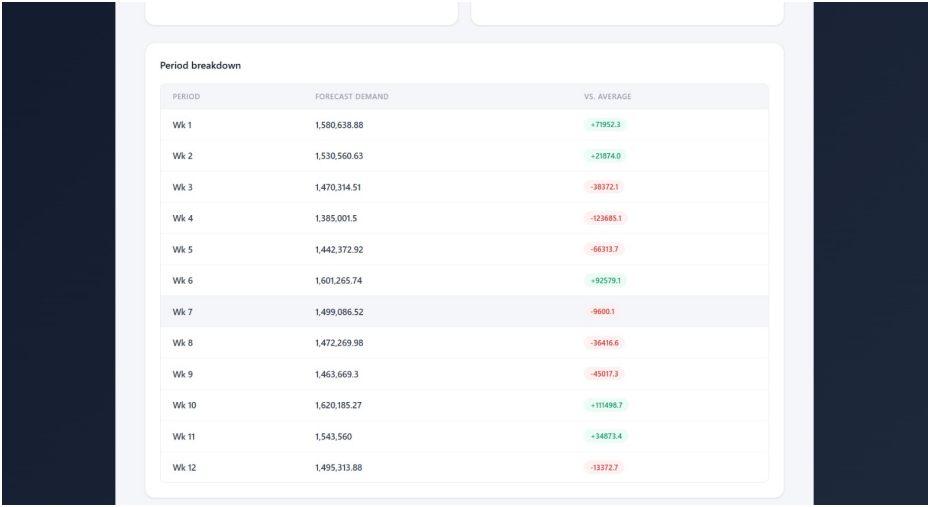


Figure 4.7: Weekly Forecast Breakdown with Deviation from Average

CHAPTER 5 : CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project demonstrates that a Sales Momentum Index-driven rule-based nudging layer can consistently improve baseline retail demand forecasts while simultaneously preserving interpretability and deployment simplicity. The framework was designed to address three well-documented weaknesses of existing retail forecasting pipelines: short-term lag in responding to demand shifts, opacity of adaptive correction mechanisms, and the high maintenance cost of frequent model retraining. By computing momentum signals directly from historical sales and applying bounded multiplier corrections as a stateless post-processing step, the SMI-Nudger framework avoids all three of these failure modes.

The experimental results across the Walmart Recruiting Store Sales and M5 Forecasting datasets yield four principal conclusions:

- First, the framework delivers substantial gains on structured, stable retail series: on the Walmart dataset, Chronos with SMI nudging achieves a 9.44% relative reduction in MASE and a 9.13% relative reduction in WQL compared to the unadjusted Chronos baseline. Prophet similarly benefits, with MASE and WQL improvements of 3.37% and 2.80% respectively. These gains are meaningful for operational retail planning, where small improvements in forecast accuracy directly translate to reductions in stockout and overstock costs.
- Second, on the sparser and more intermittent M5 dataset, the nudging framework delivers marginal but consistent improvements - between 0.12% and 0.20% relative - without degrading any model's performance. This confirms that the bounded, conservative design of the Nudger is robust: it contributes positive adjustments where momentum signals are informative and remains close to neutral where signals are noisy or ambiguous. The framework never worsens a model's performance, which is an important practical property for production deployment.
- Third, the model-agnostic design is validated empirically: the same nudging logic produces consistent directional improvements across classical decomposition models (STL-ARIMA), probabilistic time series models (Prophet), and pre-trained foundation models (Chronos) without any model-specific modifications. This generality makes the framework suitable for organizations that maintain heterogeneous forecasting model portfolios across different product categories or store formats.
- Fourth, the operational interpretability of the framework - through explicit regime labeling, bounded multipliers, and the dashboard interface - provides a practical advantage over opaque adaptive methods. Planners can understand and audit the directional adjustments applied to each forecast week, which supports organizational trust in the system and facilitates human-in-the-loop review for critical planning decisions.

Overall, the proposed SMI-Nudger framework successfully bridges the gap between static baseline forecasts and rapidly changing demand conditions through a lightweight, transparent, and model-agnostic post-processing strategy that is well-suited to practical retail deployment at scale.

5.2 Future Work

The current framework establishes a strong foundation for momentum-based forecast enhancement, and several directions for future research and engineering improvement have been identified.

- **Hierarchical SMI:** The current implementation computes momentum signals at the aggregate market level, which works well for the Walmart dataset but is less informative for the sparse item-level series in M5. Future work should investigate computing SMI signals at multiple levels of the retail hierarchy - item, store, region, and category - and aggregating or selecting the most informative level for each forecast origin. Hierarchical momentum signals would allow the Nudger to respond to local demand shifts that are invisible at the aggregate level while still leveraging market-wide momentum for items with limited individual sales history.
- **Adaptive Horizon-Aware Nudging:** The current threshold values and decay factors in the Nudger's rule table were determined through domain reasoning and validation analysis on the training folds. A natural extension is to learn these thresholds and decay parameters from validation data using a lightweight optimization procedure, potentially a grid search or Bayesian optimization over the nudge parameter space. This would allow the framework to adapt its nudge strength automatically to the characteristics of each dataset and model family, rather than relying on hand-crafted fixed thresholds.
- **Richer Exogenous Signal Integration:** The SMI framework currently derives all signals from endogenous sales history, intentionally avoiding dependence on external data sources. However, in settings where promotional calendars, weather data, or macro-economic indicators (such as CPI or unemployment) are reliably available, incorporating these signals into the confidence scoring mechanism could improve nudge calibration during atypical demand periods such as major holidays, weather events, or economic shocks.
- **Online Inference Pipeline:** The current implementation operates in a batch rolling window mode, requiring a full pipeline execution to update forecasts. A production-grade extension would implement a streaming inference architecture in which the SMI is updated incrementally as new sales data arrives (e.g., via a Kafka or Redis Streams event bus), and the Nudger re-applies corrections to the existing forecast without waiting for the next batch window. This would enable near-real-time forecast adaptation with sub-minute latency for high-frequency retail environments.
- **Business-Impact Evaluation:** The current evaluation framework quantifies forecast accuracy improvements using MASE and WQL, which are proxy metrics for operational value. A more complete evaluation would translate these forecast accuracy improvements into concrete business impact metrics - such as stockout rate reduction, overstock holding cost savings, and order quantity optimization - using a simulation of the replenishment policy driven by the nudged versus baseline forecasts. This would provide a direct monetary estimate of the framework's value to retail operators and strengthen the case for production deployment.

All of these future directions preserve the core strengths of the current framework - interpretability, modularity, and low retraining overhead - while improving robustness, adaptability, and operational relevance for large-scale retail forecasting systems.

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APPENDIX A : ACCOMPLISHED PROGRAM OUTCOMES

PO Table

PO	Attributes	How this Project Addresses the PO	Level (1/2/3)	Justification	Evidence / Reference
PO1	Engineering Knowledge	Applied time-series forecasting techniques (ARIMA, STL-ARIMA, Prophet, LSTM, Chronos) and mathematical modeling for SMI variables.	3	Demonstrated strong application of core engineering and mathematical knowledge.	Chapter 3, Chapter 4
PO2	Problem Analysis	Identified limitations of existing models such as lag, lack of interpretability, and high retraining cost.	3	Systematically analyzed a complex problem and broke it into technical challenges.	Chapter 1, Chapter 2
PO3	Design / Development of Solutions	Designed and implemented a hybrid forecasting system with a rule-based SMI Nudger and data preprocessing pipeline.	3	Developed a complete, innovative, and scalable system architecture.	Chapter 3.1
PO4	Conduct Investigations	Performed extensive experimentation using rolling window cross-validation on Walmart and M5 datasets.	3	Demonstrated strong experimental design and comparative analysis.	Chapter 4, Chapter 3.2
PO5	Modern Tool Usage	Utilized Python, Prophet, LSTM, and Chronos foundation models for implementation.	3	Used contemporary tools and advanced AI models effectively.	Chapter 3.3

PO6	The Engineer and Society	Developed a system to reduce retail product wastage, contributing to economic efficiency.	2	Addressed a real-world industrial problem with measurable impact.	Chapter 1, Chapter 5
PO7	Environment and Sustainability	Reduces overproduction and product wastage, contributing to sustainable resource utilization.	2	Supported environmental sustainability through waste reduction.	Chapter 1, Chapter 5
PO8	Ethics	Ensured data integrity (no leakage), used transparent models, and properly cited all references.	3	Demonstrated adherence to ethical standards in research.	Declaration Page, Chapter 3
PO9	Individual and Team Work	Collaboratively worked in a team to design, implement, and document the system.	3	Demonstrated effective teamwork through successful completion.	Cover Page

Table A.1: PO Mapping

APPENDIX B : SDG MAPPING

Sl. No.	SDG No. & Title	Nature of Contribution (Environmental/Social/Economic/Technological)	Direct/Indirect Contribution	Project Component/ Feature Contributing to SDG	Justification (Brief Explanation)
1	SDG 8: Decent Work And Economic Growth	Economic, Technological, Social	Direct	Retail demand forecasting pipeline (ARIMA, STL-ARIMA, Prophet, LSTM, or Chronos); SMI-Nudger decision support	The project improves forecast accuracy and short-term demand responsiveness, helping retailers optimize replenishment, reduce stock-outs/overstock, and lower operational inefficiencies, thereby supporting productivity and sustainable economic growth.

2	SDG 9: Industry, Innovation, and Infrastructure	Technological, Economic	Direct	Modular microservice architecture (frontend, backend, ML service), model-agnostic SMI post-processing layer, dashboard analytics	The system builds scalable digital forecasting infrastructure by integrating classical and foundation models with an interpretable rule-based enhancement layer, promoting innovation in data-driven retail operations.
3	SDG 12: Responsible Consumption and Production	Environmental, Economic, Social	Direct	SMI-based demand correction, 12-week forecast planning, inventory-oriented dashboard insights	The methodology reduces demand-forecast mismatch, enabling better inventory planning and minimizing waste from overproduction and unsold stock, which supports responsible resource utilization and consumption patterns.

Table B.1: SDG Mapping